ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Understanding asphalt: materials, machines and records

Trade words · clear pictures · checked explanations

Start with the words and the picture

You will trace a load through a pavement, name its layers, compare two plant processes, follow mixture through a paver, identify five screed factors, and distinguish a roller pass from coverage. You will use a picture and a short explanation together, then show your reasoning in a record another person can understand.

The figures have been redrawn from the seventh edition of *The Asphalt Handbook* (Asphalt Institute, 2007). Each figure identifies the actual book pages and any credited contributor. Your examples and calculations below are classroom inventions. They explain concepts; they do not supply a pavement design, machine setting or rolling instruction. Actual equipment work uses the identified machine's applicable manual, project requirements and qualified instruction.

Use paper, a calculator and a device that can create, save and reopen a short document. Work with a partner or instructor for the check-back activities. You may use pointing, signing, speech or another accessible communication method. Keep your own wording for writing tasks and demonstrate the actual device actions for Digital evidence.

Our fictional class is preparing an explanation of an access road to shared community facilities. Learners may choose another fictional road-use context. Personal experience and community knowledge are shared by choice; nobody needs to disclose a location or speak for a Nation. The class's explanation should be useful to a person who is new to paving.

1 · Follow the load and name the layers

Why the pavement structure matters

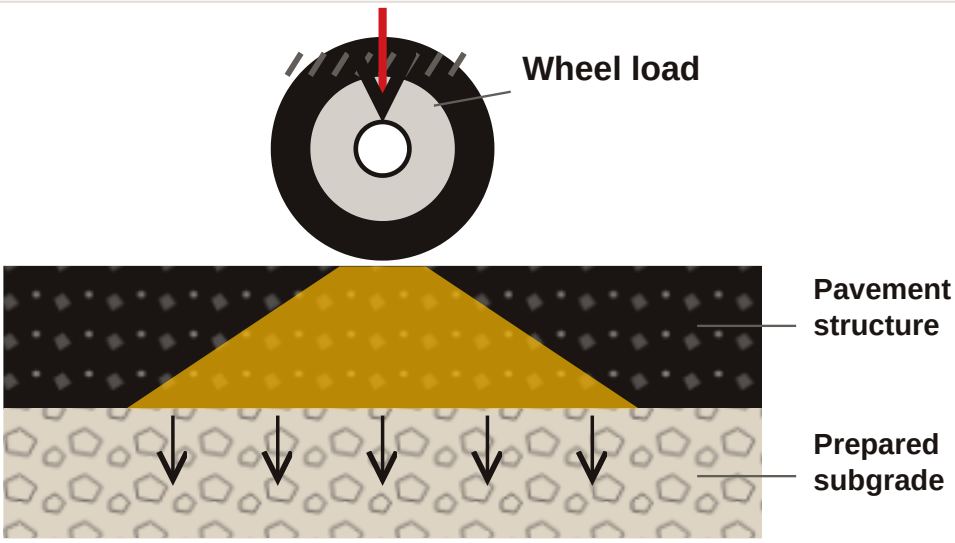
A wheel applies a load at the pavement surface. The pavement structure supports and spreads that load before it reaches the **subgrade**, the supporting ground below. The structure, subgrade and drainage must work together. A strong-looking surface alone does not establish the strength of the whole road.

The diagram is a picture of a relationship. It does not give a load value, allowable pressure or required thickness. Read the labels and direction of the arrows before making a numerical claim.

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Spread the load

Follow the load from the wheel to the supporting ground.



The pavement spreads the load before it reaches the subgrade.

Conceptual redraw · not to scale · no design thickness or pressure is specified.

Figure 1. ***Spread the load***

Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 1.5, p. 13 (PDF 47)), by Asphalt Institute, 2007. Simplified qualitative section; replaced pressure profile with a shaded load-transfer area and plain-language labels. Copyright 2007 by Asphalt Institute; original contributor credits retained.

MS4-01 · Explain the load path

Point to the wheel, pavement structure and prepared subgrade. Explain the load path in two sentences. A classmate says, “The asphalt surface carries the load, so the ground underneath does not matter.” Correct that statement using the diagram. Then name one design quantity that this drawing does **not** provide. **Skills: Reading, Writing, Communication.**

Different layers have different names

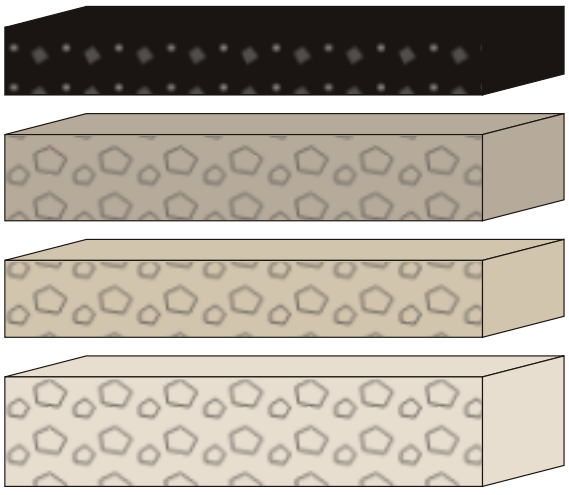
Surface course means the upper layer. **Base** and **subbase** identify supporting layers when those layers form part of the specified structure. **Prepared subgrade** identifies the prepared ground beneath the pavement structure. These terms are not interchangeable names for whatever happens to be underneath the top surface.

The second diagram shows one arrangement from the handbook: an asphalt surface above an unbound granular base and a subbase. Other arrangements in the source include full-depth asphalt and structures incorporating Portland cement concrete. Do not turn this one example into a rule that every road has four identical layers.

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Name the layers

One example: asphalt pavement on an unbound base, with a subbase.



1. **Asphalt surface**
   The upper asphalt layer.
2. **Granular base**
   Supports the asphalt surface.
3. **Subbase, where specified**
   An additional supporting layer.
4. **Prepared subgrade**
   The prepared ground beneath the structure.

Conceptual section · other pavement structures exist · layer thicknesses are not a design.

Figure 2. ***Name the layers***
Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 1.6, p. 14 (PDF 48)), by Asphalt Institute, 2007. Selected the unbound-base example, separated its layers and added functional descriptions. Other structures and design dimensions are not implied. Copyright 2007 by Asphalt Institute; original contributor credits retained.

Worked classroom example. A tabletop model card gives three layer thicknesses above the prepared ground: 40 mm surface, 100 mm base and 160 mm subbase. Their total is **40 + 100 + 160 = 300 mm = 0.300 m**. The subgrade is not a fourth thickness supplied by that card. The arithmetic totals the model dimensions; it does not approve a road design.

MS4-02 · Check a model and its limits

A different fictional model gives **50 mm surface, 120 mm base and 180 mm subbase**. Total the supplied thicknesses in millimetres and metres. Estimate first, then check the sum in a different order. Label a three-layer sketch and place the prepared subgrade beneath it. Explain why the diagram and these invented dimensions do not establish a real project's required structure. **Skills: Numeracy, Reading, Problem solving.**

2 · Distinguish batch and continuous production

Both processes proportion materials to the required mixture and combine aggregate with asphalt binder. The outline also uses **asphalt cement** for this binder. It is different from Portland cement. A batch process makes separately weighed batches. A drum/continuous process measures and combines a continuing flow. A batch plant can have a drying drum before its batch mixer; the presence of a drum alone does not identify the mixing process.

The simplified routes below omit detailed controls, recycling arrangements and additives. The source also explains that a drum/continuous facility may have a separate dryer and continuous mixer. Look for the way material is proportioned and mixed.

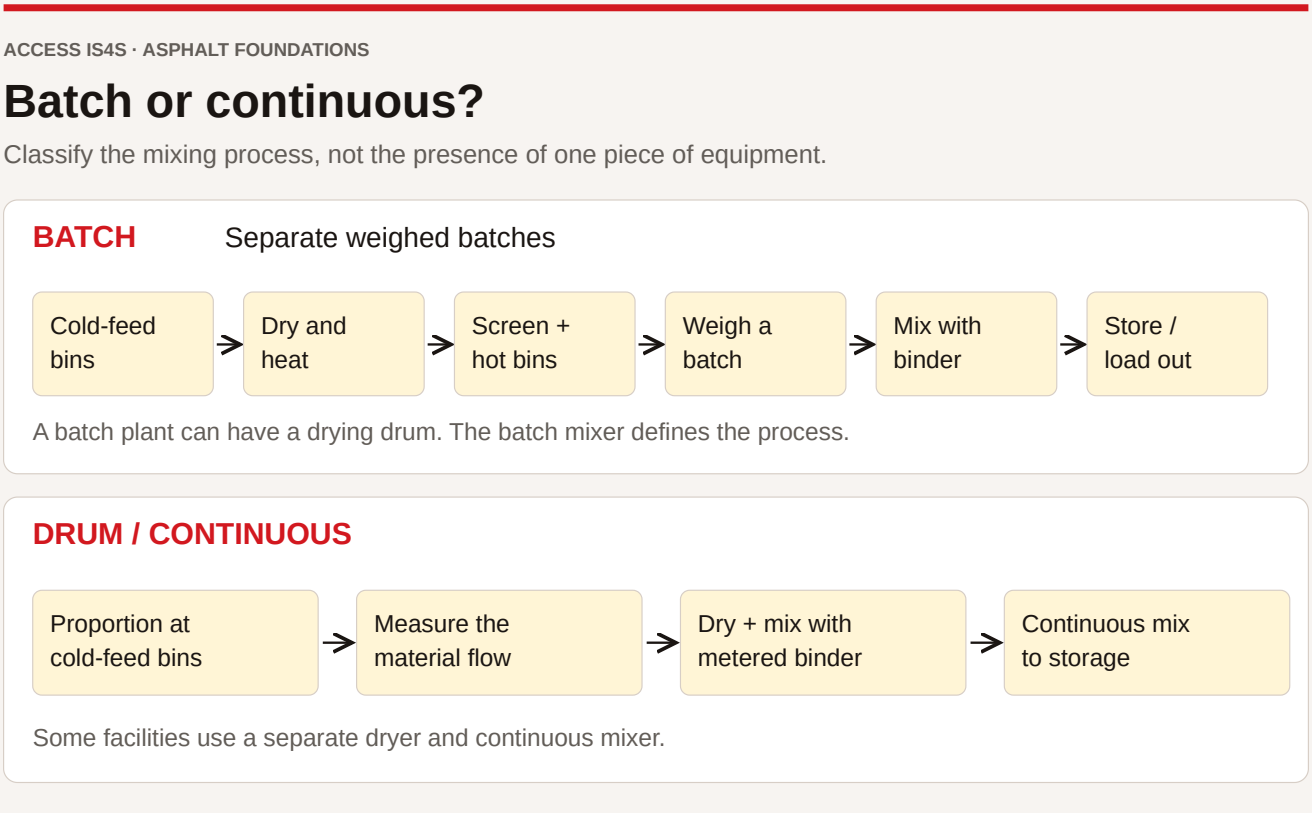


Figure 3. ***Batch or continuous?***
Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 7.2, p. 310 (PDF 344); Figure 7.4, p. 312 (PDF 346); 7.1.1–7.1.2, pp. 309, 311, 312), by Asphalt Institute, 2007. Converted equipment drawings into two simplified process lanes; omitted settings and detailed control systems. Original Figure 7.2 is credited to Terex Roadbuilding in the handbook's contributor list. Original Figure 7.4 is credited to National Highway Institute (NHI) in the handbook's contributor list. Copyright 2007 by Asphalt Institute; original contributor credits retained.

MS4-03 · Use the record, not one machine name

Read two invented production descriptions:

- **Record A:** aggregate passes through a dryer, is screened and held in hot bins, then separate quantities are weighed for a batch mixer.
- **Record B:** proportioned aggregate moves through a measured continuous feed, with binder metered into a continuing mixing process.

Name the process indicated by each record. Explain why “A has a dryer, so A must be a drum-mix plant” is not a sound conclusion. Find one function common to both routes and one difference. **Skills: Reading, Problem solving, Writing.**

MS4-04 · Make a retrievable process note

Create a short document with two labelled routes and your answer to MS4-03. Name it `asphalt-process-v1-[learning-ID]`, save it in a known practice folder, close it and reopen it. Show another person the saved location and the distinction between drying and mixing. Record the actual tool action you completed and any support. A paper draft can help you plan; keep Digital marked “not observed” until the device work is demonstrated. **Skills: Digital, Writing.**

3 · Follow the mixture through the paver

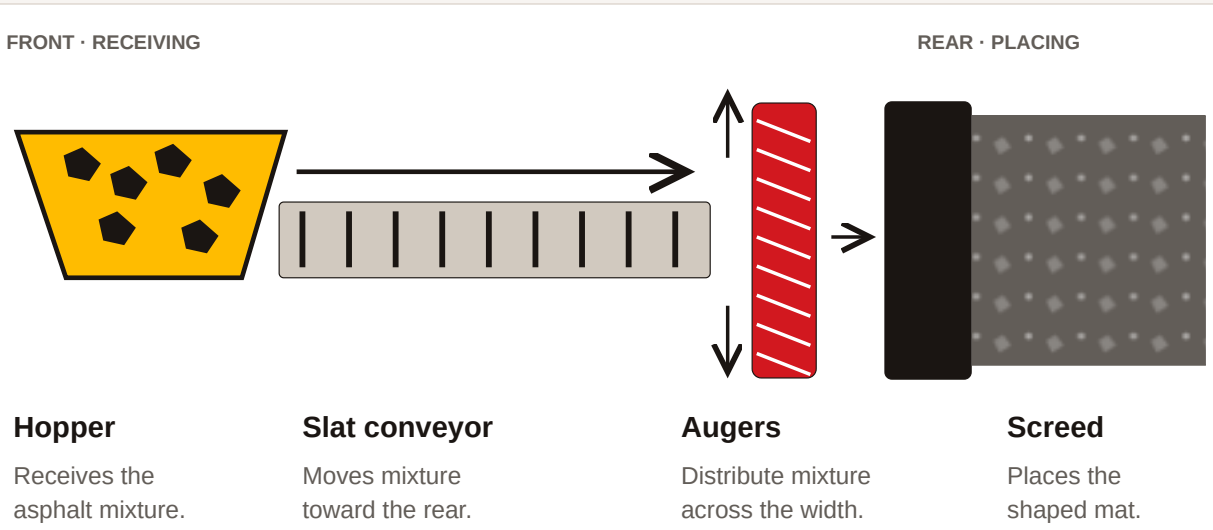
The **hopper** receives the mixture. **Slat conveyors** move it toward the rear of the paver. **Augers** distribute it across the width ahead of the screed. The **screed** places and shapes the mat. The handbook describes the paver's tractor portion as providing power, movement and material handling; the screed performs the placement function. A compaction roller has a different job after placement.

This is a functional drawing, not a picture of a particular model's controls. Compare it with the labelled LeeBoy manual figure when you need to distinguish a general function from a numbered part on one identified machine.

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Follow the mixture through the paver

Four trade words connect receiving, conveying, distributing and placing.



Functional schematic · machine geometry and controls vary · not an operating sequence.

Figure 4. ***Follow the mixture through the paver***

Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 8.11, p. 377 (PDF 411); 8.4.1, component descriptions, p. 378 (PDF 412)), by Asphalt Institute, 2007. Replaced the perspective equipment schematic with a labelled functional material route. Not drawn to scale or tied to an actual model. Original Figure 8.11 is credited to Volvo Construction Equipment in the handbook's contributor list. Copyright 2007 by Asphalt Institute; original contributor credits retained.

MS4-05 · Name the part and the function

Make four rows headed **hopper / slat conveyor / augers / screed**. Give each its function in your own words. Draw arrows to show movement toward the rear and distribution across the width. Correct this invented statement: “The augers do the same front-to-rear transporting job as the slat conveyors.” Point to the evidence for the correction. **Skills: Reading, Writing, Communication.**

MS4-06 · Check back without inventing a control

Explain one component's function to a partner. The listener restates it and asks one clarification. Swap roles with a different component. Then answer this visitor's question: “Which button should I press on our machine?” Give a useful next step using the model-specific manual and responsible instructor; the functional drawing does not identify a button. Keep an individual record of your explanation, your listener check and the source you would need next. **Skills: Communication, Collaboration, Problem solving.**

4 · Recognize a system around the screed

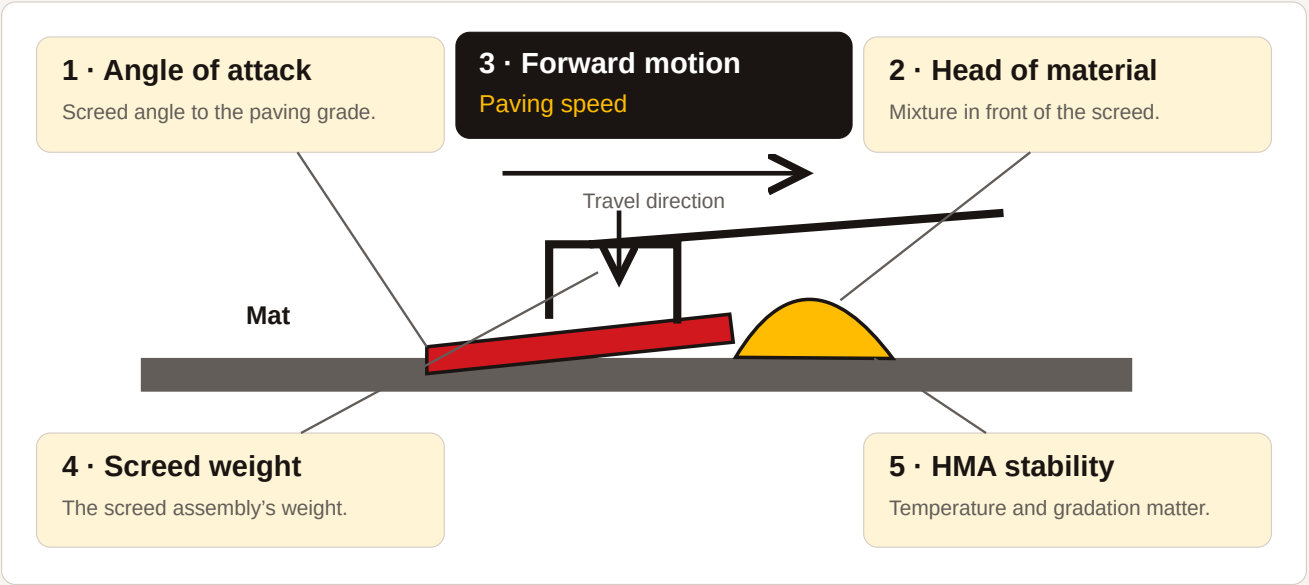
A **free-floating screed** responds to interacting conditions. MS-4 identifies five factors: **angle of attack**, **head of material**, **forward motion or paving speed**, **screed weight**, and **HMA stability**, including temperature and gradation. The **head of material** is the mixture in front of the screed. **Angle of attack** describes the screed's angle in relation to the paving grade.

Naming a factor is different from deciding how to adjust a machine. A changed result can have more than one contributing condition. First record what changed and what remains unknown, using the current machine and job information.

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A free-floating screed responds to a system

Recognize the five interacting factors before discussing a changed result.



Conceptual redraw · arrows identify relationships, not adjustment instructions or setting values.

Figure 5. ***A free-floating screed responds to a system***

Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 8.16, p. 383 (PDF 417); 8.4.3, five factors, p. 384 (PDF 418)), by Asphalt Institute, 2007. Mirrored and simplified the schematic; grouped the five factors in numbered callouts; omitted adjustment procedures and numeric settings. Original Figure 8.16 is credited to Volvo Construction Equipment in the handbook's contributor list. Copyright 2007 by Asphalt Institute; original contributor credits retained.

MS4-07 · Separate observations from assumptions

An invented classroom record states: “Paving speed changed; the amount of mixture in front of the screed also changed. A different mat result was reported. No measured thickness or temperature record was supplied.” Name the two recorded factors. List two missing facts. Explain why this record does not prove that only one factor caused the result. Do not supply an operating adjustment. **Skills: Reading, Problem solving, Writing.**

MS4-08 · Build a clearer observation record

Design a four-column form: **observation / source or measurement / time or version / information still needed**. Enter the facts from MS4-07 without turning assumptions into measurements. Ask a partner to identify one ambiguous phrase and revise it. Save or retain both versions. If using a device, reopen the revised file to check that its content changed, not just its filename. **Skills: Writing, Collaboration, Adaptability; Digital when performed.**

5 · Count a pass and describe coverage

In MS-4, a **pass** is a roller crossing over one point in the mat one time. **Coverage** concerns enough rolling to cover the complete width. These words answer different questions. A tally at one point cannot, by itself, show that the whole width has received the same rolling.

The handbook's Figure 9.1 gives one illustrated pattern. Our redraw separates the two meanings so you can read a record. Its coloured bands are a classroom model, not a prescribed roller width or operating pattern.

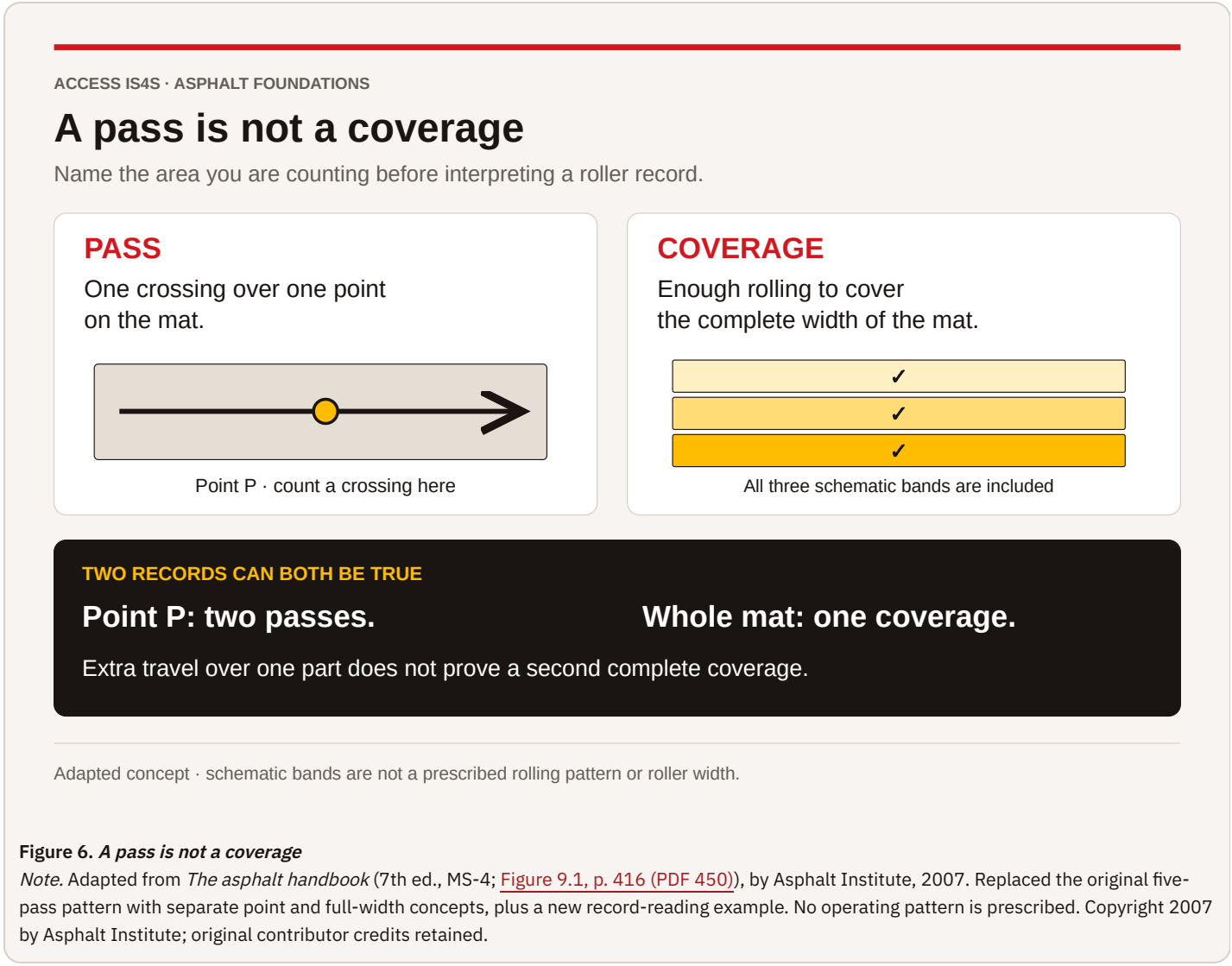


Figure 6. ***A pass is not a coverage***

Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 9.1, p. 416 (PDF 450)), by Asphalt Institute, 2007. Replaced the original five-pass pattern with separate point and full-width concepts, plus a new record-reading example. No operating pattern is prescribed. Copyright 2007 by Asphalt Institute; original contributor credits retained.

MS4-09 · Read two records precisely

The fictional log says, “Point P was crossed twice. Every part of the width was crossed at least once. Some other areas have not been crossed twice.” State the pass count at P and the number of complete coverages established by this record. Explain why two passes at P do not establish two complete coverages. **Skills: Reading, Numeracy, Problem solving.**

MS4-10 · Correct an incomplete handover

A second log says only, “The roller passed the observer three times.” Write a two-sentence handover that preserves that observation without claiming a coverage count. Ask for one piece of information needed to assess coverage. Read your handover to a partner and check what they understood. **Skills: Writing, Communication.**

6 · Make the explanation easy for another learner

MS4-11 · Build and test two explanation ideas

Choose one of the six diagrams. Design **two different** ways to help a new learner understand its main idea: for example, a labelled card sequence and a question-and-picture panel. Keep the exact trade terms and the source credit. Select one approach with a reason. Test it with a partner who has not seen your draft: ask them to find the term, explain its meaning and point to the supporting picture detail. Record what the reader actually did and one improvement you made. Each person keeps their own contribution. **Skills: Creativity and innovation, Collaboration, Reading, Writing.**

MS4-12 · Adapt the explanation for a changed brief

The visitor now needs a **black-and-white page with no more than 80 words**, excluding the figure's source note. Their question is: “What does this term mean, and what does the picture not tell me?” Keep the most useful feature of your first design and revise another. Test the revised page again. Retain the first version, changed requirement, revised version and an explanation of your choice. If you use a digital tool, save, close and reopen the new file. Colour alone must not carry the meaning. **Skills: Adaptability, Creativity and innovation, Writing; Digital when performed.**

Keep a record of the learning

Keep the activity ID, your own work, any support, one check, feedback and your next practice. You may explain a concept through an accessible communication mode. Writing evidence needs a text whose wording you control; Digital evidence needs an observed device action. A successful classroom explanation is useful evidence of learning, not an equipment endorsement.

Figure references

Asphalt Institute. (2007). *The asphalt handbook* (7th ed., MS-4). Complete local HTML reader.

The handbook credits Terex Roadbuilding for Figure 7.2; National Highway Institute for Figure 7.4; and Volvo Construction Equipment for Figures 8.11 and 8.16. These original contributor credits appear on printed pages 769–770, source PDF pages 803–804. All six ACCESS redraws identify their specific source figures and changes in the adjacent note and figure record.